

# PlantVillage Leaf Segmentation

## Detailed Methodology, Experimental Record, Challenges, and Final Results

Personal Research Notes

September 2026

### 1 Purpose of This Document

This document is a detailed personal research record for the segmentation component of a wider PlantVillage study that also contains classification and object-detection experiments. It is intentionally more detailed than a six-page conference manuscript. Its purpose is to preserve the reasoning behind preprocessing, quality-control decisions, data exclusions, model selection, training, interruption recovery, evaluation, and final results so that the complete process can later be condensed for a paper or explained during a conference presentation.

### 2 Segmentation Objective

The segmentation task is formulated as binary semantic segmentation of a plant leaf from its background. The RGB image from the PlantVillage `color` directory is used as model input. The corresponding dataset-provided image from the `segmented` directory is used as the supervision source. No independently hand-annotated mask dataset was introduced.

### 3 Dataset Audit

The working PlantVillage copy contained three parallel representations: `color`, `grayscale`, and `segmented`. A total of 38 classes were found.

The initial counts were:

- Color images: 54,305
- Grayscale images: 54,305
- Segmented images: 54,306
- Classes: 38

The segmented directory therefore contained one additional file. A class-wise audit localized the mismatch to `Grape__Esca_(Black_Measles)`. The unmatched segmented file was removed from the valid pairing set. This produced 54,305 candidate color-segmented pairs.

### 4 Color-Segmented Pairing

Equal folder counts are insufficient evidence of correct pairing; therefore, explicit one-to-one matching was performed.

The filename conventions were not fully uniform between the color and segmented directories. A two-stage procedure was used:

1. **Normalized full-name matching:** extensions and segmentation-specific filename suffixes were normalized while preserving identifying information.
2. **Descriptor-based fallback matching:** when direct normalized matching failed, a secondary descriptor key was created after normalizing identifier components and UUID-like prefixes.

The final pairing results were:

- Full-name matches: 53,109
- Descriptor fallback matches: 1,196
- Total pairs: 54,305
- Unresolved pairs: 0
- Duplicate color paths: 0
- Duplicate segmented paths: 0

Relative file paths were stored in a persistent manifest instead of Colab-specific absolute paths. This made the pairing reusable after runtime restarts and across different Google accounts, provided the same project folder structure was recreated.

## 5 Initial Train/Validation/Test Split

The 54,305 paired samples were divided using a class-stratified 80:10:10 split with random seed 42.

The initial split contained:

- Training: 43,444
- Validation: 5,430
- Test: 5,431

Sample-path intersection checks confirmed:

$$\text{Train} \cap \text{Validation} = 0, \quad \text{Train} \cap \text{Test} = 0, \quad \text{Validation} \cap \text{Test} = 0.$$

The split was persisted so that later model experiments would not recreate a new random partition.

## 6 Quality Control and Preprocessing Challenges

### 6.1 Meaning of Quality Control

Quality control (QC) refers to systematic validation performed before model training to determine whether the paired images, supervision sources, masks, image dimensions, and data partitions are valid and reproducible. QC is a data-validation process rather than a learning algorithm.

### 6.2 Spatial-Resolution Mismatches

A full scan found zero missing paired files but four color-segmented pairs with different native spatial sizes. The color inputs were  $256 \times 256$  while the segmented sources had sizes  $470 \times 512$ ,  $335 \times 512$ ,  $466 \times 512$ , and  $324 \times 512$ .

Visual inspection confirmed that these were correctly paired views of the same leaves rather than incorrect matches. They were therefore retained. When a binary target required spatial alignment, nearest-neighbor interpolation was used because it preserves discrete class labels and does not create fractional mask classes.

### 6.3 Unusable Soybean Pair

One correctly matched `Soybean_healthy` pair was unusable for segmentation. Its original image did not contain a usable leaf region and the corresponding segmented source was entirely black, with no foreground information. Since the filename pairing itself was correct, this was treated as a source-data anomaly rather than a matching failure and was excluded.

### 6.4 Non-Uniform Backgrounds and Full-Frame Leaves

The dataset-provided segmented directory was not perfectly homogeneous. Most images contained a clear near-black background, but several difficult cases were observed:

- non-black or colored backgrounds,
- leaves touching image boundaries,
- leaves occupying nearly the entire frame,
- segmented images visually close to the original color image.

A border-only QC rule initially over-flagged valid samples because a valid leaf can touch the image boundary and thereby reduce the proportion of black border pixels. The criterion was therefore refined using whole-image background statistics and visual verification.

### 6.5 Rejected Adaptive Difference + Otsu Attempt

An experimental target-generation approach was evaluated using pixel-wise color-versus-segmented RGB differences followed by Otsu thresholding. Otsu’s method automatically selects a threshold that attempts to separate two intensity populations in a histogram.

The approach worked for many images, but qualitative inspection exposed noisy and over-segmented targets for some image types. It was therefore rejected from the final pipeline. This experiment is retained in the preprocessing notebook only as a research record.

### 6.6 Final Source-Quality Screening

After QC, the 54,304 temporarily valid pairs were categorized using whole-image near-black background statistics and visual checks:

- Reliable near-black-background sources: 52,969
- True non-black candidates: 791
- Near-identical/full-frame candidates: 544

For the final segmentation experiment, only the 52,969 samples providing explicit and reliable foreground-background separation were retained. Importantly, all 38 PlantVillage classes remained represented.

The final fixed dataset was:

- Training: 42,371
- Validation: 5,300
- Test: 5,298
- Total: 52,969
- Classes: 38

No sample overlap was present between final training, validation, and test subsets.

## 7 Final Target Construction

Binary targets were generated dynamically in memory from the retained PlantVillage segmented RGB images. No separate binary-mask dataset was saved to disk.

For segmented RGB pixel  $\mathbf{x} = (R, G, B)$ :

$$M(\mathbf{x}) = \begin{cases} 1, & \max(R, G, B) > 15, \\ 0, & \max(R, G, B) \leq 15. \end{cases}$$

Thus:

- Leaf foreground = 1
- Near-black background = 0
- Threshold = 15

For the four valid size-mismatched pairs, the binary target was generated at its native resolution and then resized using nearest-neighbor interpolation.

The final target-generation method was checked over all 52,969 retained samples:

- Problematic final targets: 0
- Minimum foreground proportion: approximately 0.5005%
- Median foreground proportion: approximately 46.5332%
- Maximum foreground proportion: approximately 99.4980%

## 8 Input Preprocessing and Augmentation

All model inputs were resized to  $256 \times 256$ .

Training augmentation was applied jointly to image and target:

- horizontal flip, probability 0.5,
- vertical flip, probability 0.5,
- random 90° rotation, probability 0.5.

RGB inputs were normalized with ImageNet statistics:

$$\mu = (0.485, 0.456, 0.406), \quad \sigma = (0.229, 0.224, 0.225).$$

Validation and test images were resized and normalized without random augmentation.

## 9 Why These Three Models Were Chosen

Three segmentation architectures were evaluated to represent different design philosophies while using the same ResNet34 ImageNet-pretrained encoder.

### 9.1 U-Net

U-Net was selected as the classical encoder-decoder baseline. Its skip connections transfer high-resolution encoder information to the decoder and are well suited to recovering object boundaries and spatial detail.

## 9.2 DeepLabV3+

DeepLabV3+ was selected as a context-oriented architecture using atrous/dilated convolution and Atrous Spatial Pyramid Pooling (ASPP), followed by decoder refinement. It provides a structurally different comparison to the direct skip-connected U-Net design.

## 9.3 Feature Pyramid Network

FPN was selected to evaluate explicit multi-scale feature fusion. A top-down pathway and lateral connections combine semantically strong features from multiple spatial scales. This provides a third architectural perspective that is relevant when leaf size, shape, and occupied image area vary.

# 10 Common Loss and Optimization

The models used a combined binary cross-entropy and Dice loss:

$$\mathcal{L} = 0.5 \mathcal{L}_{BCE} + 0.5 \mathcal{L}_{Dice}.$$

Binary cross-entropy provides pixel-wise supervision, while Dice loss directly rewards foreground overlap and is useful when foreground and background frequencies are not perfectly balanced.

AdamW was used with:

- Initial learning rate:  $10^{-4}$
- Weight decay:  $10^{-4}$

A `ReduceLROnPlateau` scheduler monitored validation Dice with reduction factor 0.5 and scheduler patience 3. Mixed-precision CUDA training was used through autocasting and gradient scaling.

# 11 Evaluation Metrics and Rationale

Predicted logits were converted to probabilities with a sigmoid and binarized at threshold 0.5. Test-set confusion counts were aggregated over the complete test set before computing the final metrics.

Let  $TP$ ,  $FP$ ,  $FN$ , and  $TN$  denote pixel-level true positives, false positives, false negatives, and true negatives.

## 11.1 Dice

$$Dice = \frac{2TP}{2TP + FP + FN}.$$

Dice was used as the main overlap measure because the task directly concerns agreement between predicted and target leaf regions.

## 11.2 Intersection over Union

$$IoU = \frac{TP}{TP + FP + FN}.$$

IoU/Jaccard provides a stricter overlap criterion and is a standard semantic-segmentation measure.

## 11.3 Precision

$$Precision = \frac{TP}{TP + FP}.$$

Precision quantifies over-segmentation by measuring how many predicted foreground pixels are truly leaf pixels.

### 11.4 Recall / Sensitivity

$$Recall = \frac{TP}{TP + FN}.$$

Recall quantifies under-segmentation by measuring how much of the true leaf region is recovered.

### 11.5 Specificity

$$Specificity = \frac{TN}{TN + FP}.$$

Specificity measures how well background pixels are rejected, which is important in a leaf-versus-background task.

### 11.6 Pixel Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}.$$

Pixel accuracy provides an intuitive overall correctness measure but is not interpreted alone because class-area imbalance can make accuracy optimistic. It is therefore reported together with overlap and foreground/background error metrics.

## 12 U-Net Experiment

The U-Net experiment used:

- Encoder: ResNet34
- Encoder weights: ImageNet
- Parameters: 24,436,369
- Batch size: 16
- Maximum epochs: 30
- Early-stopping patience: 7
- Best epoch: 26

Final results:

| Metric               | U-Net    |
|----------------------|----------|
| Best validation Dice | 0.991134 |
| Best validation IoU  | 0.982472 |
| Test loss            | 0.017573 |
| Test Dice            | 0.991319 |
| Test IoU             | 0.982788 |
| Precision            | 0.991544 |
| Recall               | 0.991094 |
| Specificity          | 0.992660 |
| Pixel accuracy       | 0.991933 |

## 13 DeepLabV3+ Experiment

The DeepLabV3+ experiment used:

- Encoder: ResNet34
- Encoder weights: ImageNet
- Parameters: 22,437,457
- Batch size: 14
- Maximum epochs: 20
- Early-stopping patience: 3
- Training stopped at epoch 13
- Best epoch: 10

Final results:

| Metric               | DeepLabV3+ |
|----------------------|------------|
| Best validation Dice | 0.990076   |
| Best validation IoU  | 0.980405   |
| Test loss            | 0.020148   |
| Test Dice            | 0.990106   |
| Test IoU             | 0.980405   |
| Precision            | 0.990305   |
| Recall               | 0.989906   |
| Specificity          | 0.991584   |
| Pixel accuracy       | 0.990804   |

## 14 FPN Experiment and Recovery After Runtime Interruption

The FPN experiment used:

- Encoder: ResNet34
- Encoder weights: ImageNet
- Parameters: 23,155,393
- Batch size: 16
- Maximum epochs: 30
- Early-stopping patience: 7
- Best epoch: 25

During training, a power/runtime interruption occurred after epoch 13. The experiment was not restarted from epoch 1. Instead, training resumed from the persisted `last_fpn.pth` recovery checkpoint.

The recovery procedure restored:

- model state,
- optimizer state,

- previous training history,
- best validation Dice,
- last completed epoch,
- scheduler state when available,
- mixed-precision gradient-scaler state when available,
- early-stopping counter reconstructed from the saved validation history.

The checkpoint showed that epoch 13 had been completed. Training therefore resumed from epoch 14 with best validation Dice 0.990249, early-stopping counter 1, and learning rate  $10^{-4}$ . A recovery checkpoint was saved after every subsequent epoch. The run then completed through epoch 30, preserving a continuous 30-epoch history.

This recovery strategy is important for reproducibility because a runtime interruption did not require discarding the first 13 epochs or silently starting a different experiment.

Final FPN results:

| Metric               | FPN      |
|----------------------|----------|
| Best validation Dice | 0.990776 |
| Best validation IoU  | 0.981767 |
| Test loss            | 0.019443 |
| Test Dice            | 0.990844 |
| Test IoU             | 0.981854 |
| Precision            | 0.990803 |
| Recall               | 0.990885 |
| Specificity          | 0.992013 |
| Pixel accuracy       | 0.991488 |

## 15 Three-Model Final Comparison

| Metric               | U-Net           | DeepLabV3+ | FPN      |
|----------------------|-----------------|------------|----------|
| Best validation Dice | <b>0.991134</b> | 0.990076   | 0.990776 |
| Best validation IoU  | <b>0.982472</b> | 0.980405   | 0.981767 |
| Test loss            | <b>0.017573</b> | 0.020148   | 0.019443 |
| Test Dice            | <b>0.991319</b> | 0.990106   | 0.990844 |
| Test IoU             | <b>0.982788</b> | 0.980405   | 0.981854 |
| Precision            | <b>0.991544</b> | 0.990305   | 0.990803 |
| Recall               | <b>0.991094</b> | 0.989906   | 0.990885 |
| Specificity          | <b>0.992660</b> | 0.991584   | 0.992013 |
| Pixel accuracy       | <b>0.991933</b> | 0.990804   | 0.991488 |

The observed ranking by test Dice is:

$$\text{U-Net} > \text{FPN} > \text{DeepLabV3+}.$$

The absolute differences are small. U-Net exceeds FPN by approximately 0.000475 Dice (about 0.0475 percentage points) and DeepLabV3+ by approximately 0.001214 Dice (about 0.1214 percentage points).

These differences should be interpreted carefully because the completed experiments did not use perfectly identical training budgets: DeepLabV3+ used a smaller batch size and a shorter early-stopping budget. The same dataset, final split, preprocessing, target generation, encoder family, loss, optimizer family, initial learning rate, prediction threshold, and test protocol were maintained, but this training-budget difference is a limitation when attributing very small score differences purely to architecture.

## 16 Why the Scores Are Very High

The high performance is plausible for several reasons:

- PlantVillage imagery is more controlled than real-world field imagery.
- The final experiment deliberately retained only segmented sources with reliable foreground-background separation.
- The task is binary leaf-versus-background segmentation rather than multi-class semantic segmentation.
- All models used ImageNet-pretrained ResNet34 encoders.
- The training set is large (42,371 images).
- Training augmentation and Dice-aware optimization support stable foreground localization.

These results therefore demonstrate strong performance on the curated PlantVillage segmentation setting, but they should not be presented as proof of equal performance on unconstrained field backgrounds without external validation.

## 17 Main Challenges Encountered

The principal challenges were:

1. mismatch in folder counts (54,306 segmented images versus 54,305 color images);
2. inconsistent filename conventions across color and segmented folders;
3. initial unresolved pairing cases requiring fallback descriptor matching;
4. four valid pairs with different spatial resolutions;
5. one correctly matched but unusable blank Soybean supervision source;
6. non-uniform segmented-image backgrounds;
7. leaves touching the image border, which caused false flags under border-only QC;
8. full-frame or near-identical source images with insufficient explicit background separation;
9. failure of a simple assumption that every segmented background was uniformly black;
10. rejection of a more complex adaptive difference + Otsu approach after qualitative mask inspection;
11. runtime/power interruption during FPN training, requiring reliable checkpoint-based recovery.

## 18 Saved Reproducibility Artifacts

The preprocessing stage preserves:

- `final_segmentation_manifest.csv`,
- `segmentation_preprocessing_config.json`,
- QC exclusion records,
- pairing manifests and audit outputs.

Each model experiment preserves:

- best and last checkpoints,
- configuration JSON,
- training-history CSV,
- test-metrics JSON/CSV,
- experiment-summary JSON,
- loss, Dice, and IoU curves,
- qualitative prediction figures.

Large .pth checkpoints are retained in Google Drive/local backup rather than normal Git tracking.

## 19 Conference-Presentation Q&A Notes

### **Why were approximately 1,335 samples excluded after the initial pairing stage?**

The final dataset was not reduced simply to improve performance. Samples were removed because the segmentation supervision source did not provide sufficiently reliable explicit foreground-background separation under the finalized target-generation rule. The objective was to avoid training on ambiguous pseudo-ground truth. Importantly, all 38 classes remained represented after filtering.

### **Why not generate new masks manually?**

The study intentionally used the segmented representation already supplied by PlantVillage as the supervision source. The goal was to avoid introducing a separately annotated mask dataset and to keep the segmentation task directly linked to the original dataset.

### **Why nearest-neighbor interpolation for masks?**

Binary masks contain discrete labels. Bilinear or bicubic interpolation can create intermediate fractional values near boundaries, whereas nearest-neighbor interpolation preserves the class labels 0 and 1.

### **Why use both Dice and IoU?**

Dice provides an intuitive overlap measure and is commonly optimized in segmentation, whereas IoU is stricter. Reporting both prevents reliance on a single overlap statistic.

### **Why report precision, recall, specificity, and accuracy as well?**

Precision reveals excessive foreground prediction, recall reveals missed leaf regions, specificity measures background rejection, and accuracy provides overall pixel correctness. Together with Dice and IoU they explain the type of segmentation error, not only the amount of overlap.

### **Why did U-Net perform slightly better?**

The observed results show that U-Net achieved the highest scores on this dataset, but the score differences are small. The controlled PlantVillage backgrounds and strong skip-connected spatial reconstruction may favor U-Net, but the experiments do not justify claiming that U-Net is universally superior.

## Did the FPN interruption invalidate the run?

No. The last completed checkpoint was saved after each epoch. After the interruption, the model, optimizer, training history, best score, scheduler/scaler state where available, and early-stopping state were restored. Training resumed from epoch 14 after epoch 13 had already completed, rather than restarting from the beginning.

## 20 Conference-Level Assessment of the Segmentation Component

From an experimental-execution perspective, the segmentation component is conference-grade in several respects: it includes explicit data auditing, reproducible pairing, a frozen stratified split, leakage checks, documented exclusions, fixed preprocessing, multiple established architectures, best-checkpoint selection, complete test-set evaluation, qualitative predictions, and persistent experiment artifacts.

However, experimental quality is not the same as research novelty. PlantVillage is a controlled and widely used dataset, and U-Net, DeepLabV3+, and FPN are established architectures. Therefore, the eventual ICCIT contribution should not be framed merely as “three segmentation models on PlantVillage.” The stronger paper-level contribution is likely to depend on the complete unified classification–segmentation–detection framework, the consistency of the experimental protocol, any transfer-learning design used in detection, the cross-task analysis, and how clearly the paper demonstrates a research question beyond simple benchmark comparison.

A final judgment about ICCIT 2026 suitability should therefore be made after the classification and detection implementations and complete manuscript contribution are reviewed.

## 21 References

### References

- [1] S. P. Mohanty, D. P. Hughes, and M. Salathé, “Using deep learning for image-based plant disease detection,” *Frontiers in Plant Science*, vol. 7, Art. no. 1419, 2016.
- [2] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional Networks for Biomedical Image Segmentation,” in *MICCAI*, 2015.
- [3] K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” in *CVPR*, 2016.
- [4] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, “Encoder-Decoder with Atrous Separable Convolution for Semantic Image Segmentation,” in *ECCV*, 2018.
- [5] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, “Feature Pyramid Networks for Object Detection,” in *CVPR*, 2017.
- [6] I. Loshchilov and F. Hutter, “Decoupled Weight Decay Regularization,” in *ICLR*, 2019.